# Veronica Boyce

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## EDUCATION

***Stanford University*** **Stanford, CA**
*Doctor of Philosophy in Psychology* September 2025
Dissertation: *Constraints on the emergence and interpretability of linguistic conventions*
***Massachusetts Institute of Technology*** **Cambridge, MA**
*Bachelor of Science in Brain and Cognitive Sciences* June 2019

## EMPLOYMENT

***MIT, Computational Psycholinguistics Lab*** **Cambridge, MA**
*NSF SBE Postdoctoral Research Fellow* Jan 2026 – present
***Stanford University, Language and Cognition Lab*** **Stanford, CA**
*Postdoctoral Researcher* Sep 2025 – Jan 2026
***MIT, Computational Psycholinguistics Lab*** **Cambridge, MA**
*Research Support Associate* June 2019 – July 2020

## AWARDS, FELLOWSHIPS, & GRANTS

NSF SBE Postdoctoral Research Fellowship Jan 2026 – Jan 2028
Open Science Innovator Prize, *Stanford Center for Open and Reproducible Science* 2025
The Albert H. and Barbara R. Hastorf Prize for Teaching, *Stanford Department of Psychology* 2025

## PUBLICATIONS

**V. Boyce**, R. Z. Sparks, and M. C. Frank (2026). "Preschoolers can coordinate with each other to communicate about novel referents". *Open Mind*. doi.org/10.1162/opmi.a.399.

M. C. Frank, V. A. Marchman, C. A. Bergey, **V. Boyce**, M. Braginsky, G. Kachergis, J. Mankewitz, S. C. Meylan, B. Prystawski, N. Ram, R. Z. Sparks, A. Steffan, A. W. M. Tan, and M. Zettersten (2026). "Continuous developmental changes in word recognition support language learning across early childhood". *eLife*. elifesciences.org/articles/109636.

M. Soderstrom, J. Rocha-Hidalgo, L. E. Muñoz, A. Bochynska, J. F. Werker, B. Skarabela, A. Seidl, Y. Ryjova, J. L. Rennels, C. E. Potter, M. Paulus, M. Ota, N. M. Olesen, K. M. Nave, J. Mayor, A. Martin, L. C. Machon, C. Lew-Williams, E.-S. Ko, H. Kim, N. Kartushina, M. Kammermeier, A. Jessop, J. F. Hay, N. Havron, E. E. Hannon, J. K. Hamlin, N. Gonzalez-Gomez, A. Gampe, T. Fritzsche, M. C. Frank, S. Durrant, C. Davies, C. Cashon, K. Byers-Heinlein, **V. Boyce**, A. K. Black, C. Bergmann, L. Anderson, M. K. Alshakhori, A. H. Al-Hoorie, and A. S. M. Tsui (2025). "Testing the relationship between preferences for infant-directed speech and vocabulary development: A multi-lab study". *Journal of Child Language*. doi.org/10.1017/S0305000924000254.

**V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2024). "Interaction Structure Constrains the Emergence of Conventions in Group Communication". *PNAS*. doi.org/10.1073/pnas.2403888121.

**V. Boyce**, B. Prystawski, A. B. Abutto, E. M. Chen, Z. Chen, H. Chiu, I. Ergin, A. Gupta, C. Hu, B. Kemmann, N. Klevak, V. Y. Q. Lua, M. Mazzaferro, K. Mon, D. Ogunbamowo, A. Pereira, J. Troutman, S. Tung, R. Uricher, and M. C. Frank (2024). "Estimating the replicability of psychology experiments after an initial failure to replicate". *Collabra: Psychology*. doi.org/10.1525/collabra.125685.

A. Chuey, **V. Boyce**, A. Cao, and M. C. Frank (2024). "Conducting Developmental Research Online vs. in-Person: A Meta-Analysis". *Open Mind*. doi.org/10.1162/opmi_a_00147.

**V. Boyce** and R. P. Levy (2023). "A-Maze of Natural Stories: Comprehension and Surprisal in the Maze Task". *Glossa Psycholinguistics*. escholarship.org/uc/item/6vh9d8zm.

**V. Boyce**, M. Mathur, and M. C. Frank (2023). "Eleven Years of Student Replication Projects Provide Evidence on the Correlates of Replicability in Psychology". *Royal Society Open Science*. doi.org/10.1098/rsos.231240.

M. Zettersten, D. Yurovsky, T. L. Xu, S. Uner, A. S. M. Tsui, R. M. Schneider, A. N. Saleh, S. C. Meylan, V. A. Marchman, J. Mankewitz, K. MacDonald, B. Long, M. Lewis, G. Kachergis, K. Handa, B. deMayo, A. Carstensen, M. Braginsky, **V. Boyce**, N. S. Bhatt, C. A. Bergey, and M. C. Frank (2023). "Peekbank: An Open, Large-Scale Repository for Developmental Eye-Tracking Data of Children's Word Recognition". *Behavior Research Methods*. doi.org/10.3758/s13428-022-01906-4.

**V. Boyce**, R. Futrell, and R. P. Levy (2020). "Maze Made Easy: Better and Easier Measurement of Incremental Processing Difficulty". *Journal of Memory and Language*. doi.org/10.1016/j.jml.2019.104082.

## CONFERENCE PROCEEDINGS

A. W. M. Tan, B. Prystawski, and **V. Boyce** (2026). "Which one is banana man? Evaluating vision-language models in multi-turn pragmatic interpretation". *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*. Talk, aclanthology.org/2026.emnlp-main.548.

**V. Boyce**, B. Prystawski, A. W. M. Tan, and M. C. Frank (2025). "Idiosyncratic but not opaque: Linguistic conventions formed in reference games are interpretable by naïve humans and vision–language models". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, escholarship.org/uc/item/16c4n85d.

**V. Boyce**, R. Z. Sparks, Y. Mofor, and M. C. Frank (2025). "Preschoolers can form conventional pacts with each other to communicate about novel referents". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, osf.io/dpgfw_v1.

A. W. M. Tan, B. Prystawski, **V. Boyce**, and M. C. Frank (2025). "Context informs pragmatic interpretation in vision-language models". *CogInterp workshop NeurIPS 2025*. Poster, preprint at arxiv.org/abs/2511.03908.

**V. Boyce** and M. C. Frank (2023). "Communicative Reduction in Referring Expressions within a Multi-Player Negotiation Game". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, escholarship.org/uc/item/8dq8c2s6.

**V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2022). "Two's Company but Six Is a Crowd: Emergence of Conventions in Multiparty Communication Games". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Talk, osf.io/preprints/psyarxiv/xd7ep.

J. Mankewitz, **V. Boyce**, B. Waldon, G. Loukatou, D. Yu, J. Mu, N. D. Goodman, and M. C. Frank (2021). "Multi-Party Referential Communication in Complex Strategic Games". *Meaning in Context: Pragmatic Communication in Humans and Machines workshop. NeurIPS.* PsyArXiv. Talk, osf.io/preprints/psyarxiv/tfb3d.

M. Zettersten, C. A. Bergey, N. S. Bhatt, **V. Boyce**, M. Braginsky, A. Carstensen, B. deMayo, G. Kachergis, M. Lewis, B. Long, K. MacDonald, J. Mankewitz, S. C. Meylan, A. N. Saleh, R. M. Schneider, A. S. M. Tsui, S. Uner, T. L. Xu, D. Yurovsky, and M. C. Frank (2021). "Peekbank: Exploring Children's Word Recognition through an Open, Large-Scale Repository for Developmental Eye-Tracking Data". *Proceedings of the Annual Meeting of the Cognitive Science Society.* Poster, escholarship.org/uc/item/9j05h4n1.

**V. Boyce**, A. Aravind, and M. Hackl (2017). "Lexical and Syntactic Effects on Auxiliary Selection: Evidence from Child French". *Boston University Conference on Language Development.* Talk, lingref.com/bucld/41/BUCLD41-08.pdf.

## PREPRINTS (UNDER REVIEW)

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**V. Boyce**, V. A. Marchman, H. A. Baumgartner, C. A. Bergey, M. Braginsky, F. Giovannetti, G. Kachergis, J. Mankewitz, S. C. Meylan, B. Prystawski, R. Z. Sparks, A. Steffan, A. W. M. Tan, M. C. Frank, and M. Zettersten (2026). "Data-driven recommendations for increasing reliability and validity in measures of infant word recognition". Preprint, under review, osf.io/preprints/psyarxiv/28gq4_v2.

**V. Boyce**, A. W. M. Tan, J. Mankewitz, F. Feng, B. Fernandez, R. Fried, M. C. Frank, and R. D. Hawkins (2026). "Refbank: an open repository of iterated reference game data". Preprint, under review, osf.io/preprints/psyarxiv/d93gy_v1.

A. N. Meltzoff, S. Farwaha, V. Slaughter, E. A. Simpson, R. Brooks, M. Nielsen, H. A. Baumgartner, L. Bayet, S. Benavides-Varela, **V. Boyce**, K. Byers-Heinlein, J. Dönicke, B. P. Flaherty, S. H. Forbes, M. C. Frank, A. Geraci, K. Hamlin, M. Heimann, N. Keven, J. Lany, J. R. Lelonkiewicz, T. S. Leung, S. Maylott, D. Moreau, E. Quadrelli, P. J. Reschke, C. Turati, S. Vincini, S. M. Weiss, T. S. Zamuner, and G. Zeng (2026). "ManyBabies 6: A Multi-Lab Investigation of Neonatal and Early Imitation". *Developmental Science.* Stage 1 Registered Report, in-principle acceptance, preprint, osf.io/preprints/psyarxiv/fe7pj_v1.

**V. Boyce**, A. Acharya, and M. C. Frank (2025). "Use of communication channels in 2x2 strategic games". White paper, osf.io/preprints/psyarxiv/y6ahw_v1.

L. Röseler, L. Wallrich, H. Hartmann, L. Altegoer, **V. Boyce**, S. M. Field, J. Goltermann, J. Hüffmeier, C. R. Pennington, M.-M. Pittelkow, P. Silverstein, D. van Ravenzwaaij, and F. Azevedo (2025). "Handbook on Conducting Reproduction and Replication Studies". Preprint, under revision following MetaROR peer review, doi.org/10.5281/zenodo.16990114.

## CONFERENCE PRESENTATIONS

**V. Boyce**, A. W. M. Tan, J. Mankewitz, F. Feng, B. Fernandez, R. Fried, B. Prystawski, M. C. Frank, and R. D. Hawkins (2026). "Refbank: an open repository of iterated reference games". The 39th Annual Conference on Human Sentence Processing. Poster.

**V. Boyce** and R. P. Levy (2025). "Linguistic, but not visual, discourse context modulates syntactic parsing in the Maze". The 38th Annual Conference on Human Sentence Processing. Poster.

**V. Boyce**, B. Prystawski, A. W. M. Tan, and M. C. Frank (2025). "Context matters: History and informativity affect understanding of opaque references". The 38th Annual Conference on Human Sentence Processing. Poster.

J. Vigly, **V. Boyce**, M. Socolof, J. Michaelov, T. von der Malsburg, and R. P. Levy (2025). "Implicit gender bias in linguistic descriptions for expected events: The case of the 2024 United States presidential election". The 38th Annual Conference on Human Sentence Processing. Poster.

**V. Boyce**, I. Chen, R. Z. Sparks, M. Perez, and M. C. Frank (2024). "4-5 year old children can successfully communicate using ad-hoc referential expressions". CAMP 6. Talk.

**V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2023). "Thick Feedback Facilitates Referential Coordination in Larger Groups". CAMP 5. Talk.

**V. Boyce**, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2023). "Thick Feedback Facilitates Referential Coordination in Larger Groups". HSP. Poster.

T. Poppels, **V. Boyce**, C. Ajunwa, T. von der Malsburg, and R. P. Levy (2021). "Bias against 'she' Pronouns Can Be Rapidly Overcome by Changing Event Expectations". CUNY Human Sentence Processing Conference. Short talk.

**V. Boyce** and R. P. Levy (2020). "A-Maze of Natural Stories: Texts Are Comprehensible Using the Maze Task". Architectures and Mechanisms for Language Processing. Talk.

**V. Boyce**, R. Futrell, and R. P. Levy (2019). "Maze Made Easy: Better and Easier Measurement of Incremental Processing Difficulty". CUNY Human Sentence Processing Conference. Poster.

**V. Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2019). "Female Gender Is Consistently Under-Expressed in Pronoun Production and Under-Inferred in Comprehension". Linguistic Society of America Annual Meeting. Poster.

**V. Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2019). "Remember 'Him', Forget 'Her': Gender Bias in the Comprehension of Pronominal Referents." CUNY Human Sentence Processing Conference. Poster.

**V. Boyce**, T. von der Malsburg, T. Poppels, and R. P. Levy (2018). "Implicit Gender Biases in the Production and Comprehension of Pronominal References". CUNY Human Sentence Processing Conference. Poster.

## INVITED TALKS

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| Harvard LangCog Talk Series, Harvard University | Sep 2026 |
| Goldenberg Lab, Harvard Business School | Nov 2025 |
| Language and Cognitive Development Lab, UC Berkeley | Oct 2025 |
| Saxelab, MIT | Feb 2023 |
| QuantLang Lab, UC Irvine | April 2021 |

## TEACHING

***Stanford, Department of Psychology*** **Stanford, CA**

*TA for PSYCH 125: Research Methods in Psychology* Jan 2024 – March 2024
*First offering of course; designed and led course replication projects with bi-weekly milestones*

*TA for PSYCH 251: Experimental Methods* Sep 2023 – Dec 2023
*Designed and supervised re-replication option for course project*
*Led collective writing of joint paper reporting findings across projects (see Publications)*

*TA for PSYCH 251: Experimental Methods* Sep 2022 – Dec 2022

*Guest Lecturer, PSYCH 251: Experimental Methods* 2023, 2025
*Topic: meta-analysis and meta-science*

*TA for PSYCH 45: Introduction to Learning and Memory* April 2023 – June 2023

*TA for PSYCH 220B: Probabilistic Models of Cognition: Language* Sep 2021 – Dec 2021

***MIT, Department of Brain and Cognitive Sciences*** **Cambridge, MA**

*TA for Computational Psycholinguistics* February 2019 – May 2019

## MENTORING

Benji Fernandez (Stanford, undergraduate honors thesis in progress, co-advisor) 2025 – 2026
*Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints*

Fiona Feng (Queens College undergraduate research assistant) 2025 – 2026
*Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints*

Ryder Fried (Stanford undergraduate research assistant) 2025
*Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints*

Spurti Nimbali (Stanford undergraduate research assistant) 2025

Zihan Wang (Stanford undergraduate research assistant) 2025

Yannick Mofor (Stanford undergraduate research assistant) 2024
*Co-author, CogSci proceedings, see Conference Proceedings*

Rebecca Pizzitola (Stanford undergraduate research assistant) 2024

Ilaria Chen (Stanford undergraduate research assistant) 2023
*Co-author, talk at CAMP, see Conference Presentations*

Malia Perez (Stanford undergraduate research assistant) 2023
*Co-author, talk at CAMP, see Conference Presentations*
*Now PhD student at UNC Chapel Hill*

Brittany Linus (Stanford undergraduate research assistant) 2022

Chelsea Ajunwa (MIT undergraduate research assistant) 2020
*Co-author, short talk at HSP, see Conference Presentations*
*Now PhD student at Northeastern University*

Arun Wongprommoon (MIT undergraduate research assistant) 2019

## AD HOC REVIEWING

*Journals:* Replication Research (2026); Journal of Personality and Social Psychology: Attitudes and Social Cognition (2026); Open Mind (2025–2026); Infant Behavior and Development (2026); Language, Cognition, and Neuroscience (2024); Royal Society Open Science (2024); Journal of Experimental Psychology: Learning, Memory, and Cognition (2023); Journal of Memory and Lan-

guage (2023); Glossa Psycholinguistics (2021)
*Co-reviewed (with Michael C. Frank):* Psychological Science (2025), Perspectives on Psychological Science (2025), Nature Human Behaviour (2024), Glossa Psycholinguistics (2023)
*Conferences:* CogSci (2021–2026), HSP (2026), California Meeting of Psycholinguists (2023), Society for Computation in Linguistics (2020–2026), Association for Computational Linguistics (2019)

## REFERENCES

Dr. Roger Levy, Department of Brain and Cognitive Sciences, MIT
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Dr. Michael C. Frank, Department of Psychology, Stanford University
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Dr. Robert Hawkins, Department of Linguistics, Stanford University
rdhawkins@stanford.edu, (217) 549-6923